

Q1 Sum the given number upto single digit

5 marks

Write a Python code to find the sum of the given number upto single digit

input: 89789

output : 5

Your Code

```
1 num=int(input())
2 while num>=10:
3     s=0
4     while num>0:
5         s+=num%10
6         num//=10
7     num=s
8 print(num)
```

Input

89789

Output

5

1.

Q2 Prime Number

5 marks

Find the next prime number

input: 13

output: 17

Your Code

```
1 num=int(input())
2 num=num+1
3 while True:
4     prime=True
5     for i in range(2,num):
6         if num%i==0:
7             prime=False
8             break
9     if prime:
10        print(num)
11        break
```

Input

13

Output

17

2.

Q3 Swap the first and last digits

5 marks

WAP to swap the first and last digits of the given number.

Input: 23546

Output: 63542

Your Code

```
1 num=input()
2 if len(num)==1:
3     print(num)
4 else:
5     new_num=num[-1]+num[1:-1]+num[0]
6     print(new_num)
```

Input

23546

Output

63542

3.

4.

Q4 Odd Sequence first and Even Sequence next

WAP to odd Sequence first and Even Sequence next

Input: 658134

Output: 513684

Your Code

```
1 num=input()
2
3 odd=""
4 even=""
5
6 for digit in num:
7     if int(digit)%2!=0:
8         odd+=digit
9     else:
10        even +=digit
11 print(odd+even)
```

Input

658134

Output

513684

Q5 Step Number

WAP to check the given number is step number or not

A step number is a number in which the difference between every pair of consecutive digits is 1.

For example:

54567 → Yes

$|5-4| = 1$

$|4-5| = 1$

$|5-6| = 1$

$|6-7| = 1$

1235 → No

$|3-5| = 2$

5.

Your Code

```
1 n=input("")
2
3 flag=1
4 for i in range(len(n)-1):
5     if abs(int(n[i])-int(n[i+1]))!=1:
6         flag=0
7         break
8 if flag==1:
9     print("Yes")
10 else:
11     print("No")
```

Input

54567

Output

Yes

Visible Test Cases

6.

Q6 Reverse only the even numbers

5 marks

WAP to reverse only the even numbers

Input: 325649

Output: 345629

Your Code

```
1 num=input()
2
3 even=[]
4 for ch in num:
5     if int(ch)%2==0:
6         even.append(ch)
7 even.reverse()
8 result=""
9 j=0
10 for ch in num:
11     if int(ch)%2==0:
12         result+=even[j]
13         j+=1
14     else:
15         result+=ch
16 print(result)
```

Input

325649

Output

345629

7,

Q7 Encoding 5 marks

WAP to encode the given number
Input: 55553322255
Output: 54322352

Your Code

```
1 n=input()
2 count=1
3 ans=""
4 for i in range(len(n)):
5     if i==len(n)-1:
6         ans+=n[i]+str(count)
7     elif n[i]==n[i+1]:
8         count+=1
9     else:
10        ans+=n[i]+str(count)
11        count=1
12
13 print(ans)
```

Input
55553322255

Output
54322352

8.

Q8 Reverse Odd Digit 5 marks

WAP to Reverse Odd Digits Only
Input: 816253
Output: 351

Your Code

```
1 n=input()
2 for i in range(len(n)-1,-1,-1):
3     if int(n[i])%2!=0:
4         print(n[i],end="")
```

Input
816253

Output
351

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9.

Q9 Convert every number into the next sequence number.

5 marks

WAP to convert every number into the next sequence number.

Input: 89781

Output: 90892

Your Code

```
1 num=input()
2
3 result=""
4
5 for ch in num:
6     if ch=='9':
7         result+='0'
8     else:
9         result+=str(int(ch)+1)
10 print(result)
```

Input

89781

Output

90892

10.

WAP to print the total factor count of a given number

Input: 12

Output: 6

Your Code

```
1 num=int(input())
2 count=0
3 for i in range(1,num+1):
4     if num%i==0:
5         count+=1
6 print(count)
```

Input

12

Output

6

Visible Test Cases

Test Case 1



5.00 / 5 marks

Submitted